

SMART INDIA HACKATHON 2025

EduMitra AI



Rajasthan AI Student Assistance Chatbot

Problem Statement ID - [SIH2025-EDTECH-03]

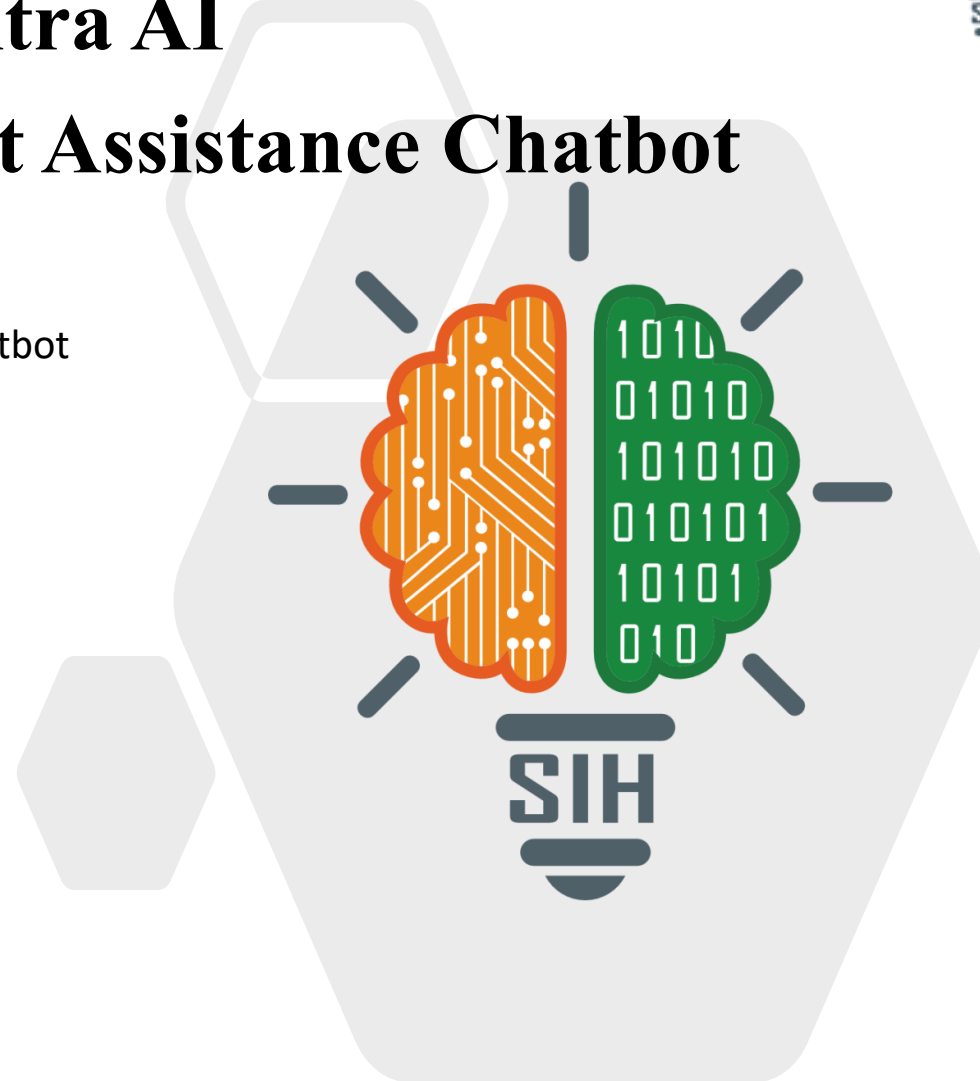
Problem Statement Title - Rajasthan AI Student Assistance Chatbot

Theme - Smart Education / EdTech

PS Category - Software

Team ID - [Your Team ID]

Team Name - Endeavours



â€¢ **Multilingual AI Chatbot (Hindi / English / Voice):** Answers natural language queries regarding REAP/DTE cutoffs, fees, hostels, eligibility and scholarships.

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â€¢ **Smart College Recommendation Engine:** Suggests optimal engineering and polytechnic colleges based on student REAP rank, category, branch, and budget.

â€¢ **Side-by-Side College Comparison:** Evaluates institutions across placements, infrastructure, NAAC grade, and fee structures.

â€¢ **Scholarship and Welfare Finder:** Automatically maps eligible state (CM Higher Education) and central scholarships.

â€¢ **80%+ Reduction in Staff Workload:** Automates repetitive admission office inquiries for all Rajasthan technical institutes.



EduMitra AI - Centralized AI Student Assistant for Rajasthan Admissions

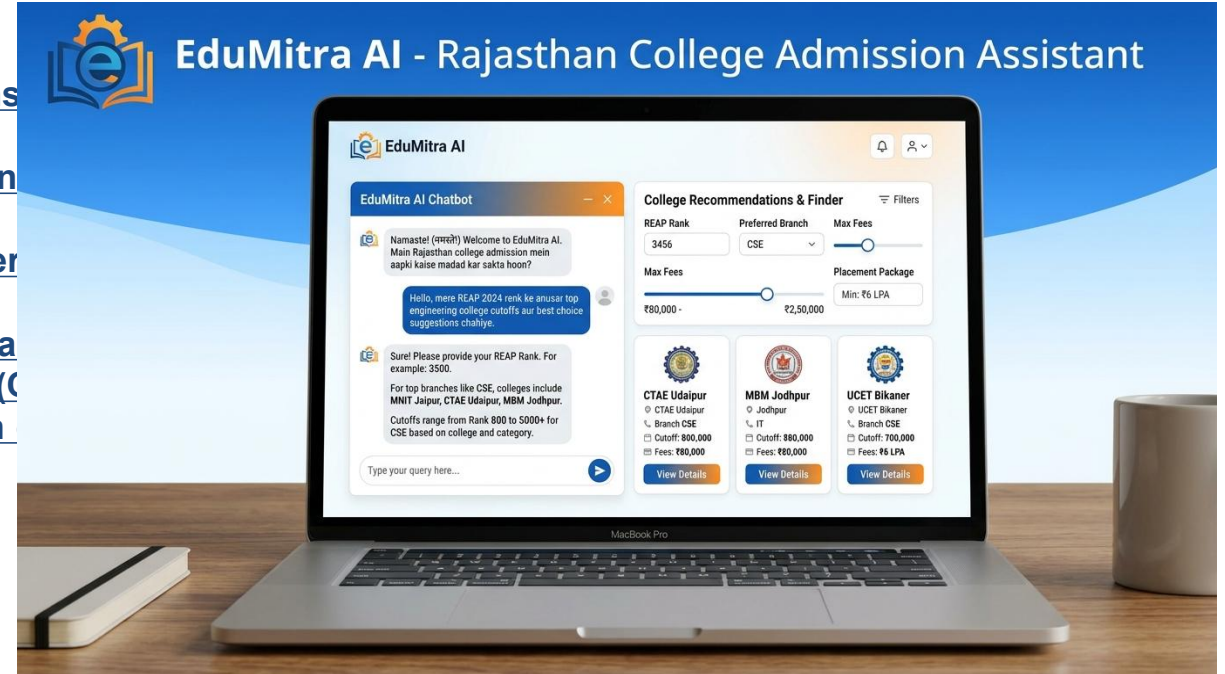
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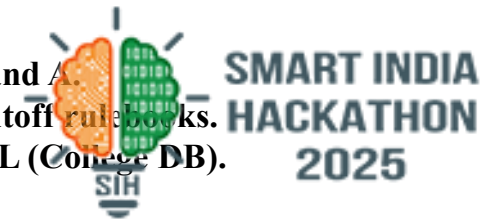
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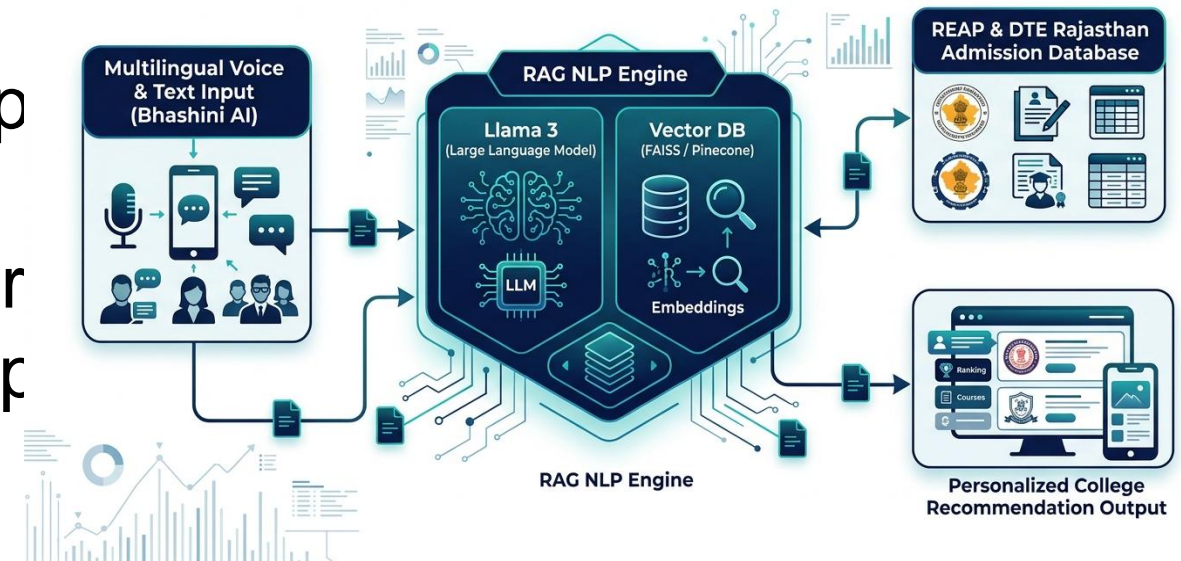
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â€¢ Multilingual NLP and Voice Core: Bhashini API + Llama 3 for real-time Hindi/English voice and text Q and A.
 â€¢ RAG (Retrieval-Augmented Generation): Vector DB (ChromaDB) indexing official REAP/DTE seat matrix and cutoff rule books.
 â€¢ Tech Stack: React.js (Frontend PWA), Python FastAPI (AI Microservices), Node.js (Backend Gateway), PostgreSQL (College DB).
 â€¢ High Accuracy and Zero Hallucination: Strict RAG grounding with source document citations.



- Technologies to be used (e.g. p frameworks, hardware)
- Methodology and process for Charts/Images/ working prototyp

EduMitra AI: ADMISSION RECOMMENDATION SYSTEM ARCHITECTURE



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FEASIBILITY AND PRACTICALITY:

â€¢ High Technical Feasibility: Built on open-source Llama 3 + RAG requiring lightweight server infrastructure.

â€¢ High Accessibility: Works via Web PWA and WhatsApp Bot for low-bandwidth rural access across Rajasthan.



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2025

RISKS AND STRATEGIES:

â€¢ Challenge: Dynamic yearly REAP cutoff and seat updates -> Strategy: Automated admin verification pipeline for DTE data.

â€¢ Challenge: Rural accessibility and low digital literacy -> Strategy: Voice-based search in Hindi and regional dialects.

â€¢ Challenge: Information Accuracy -> Strategy: Verified source linking to official DTE Rajasthan portals.

- Analysis of the feasibility of the idea
- Potential challenges and risks
- Strategies for overcoming these challenges

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IMPACT AND BENEFITS:

- â€¢ For Students and Parents: Transparent, instant, 24/7 admission guidance without travel or middleman fees.
- â€¢ For Colleges and Staff: 80% decrease in repetitive inquiry calls/emails; automated query routing.
- â€¢ For Govt and DTE Rajasthan: Digitized student counselling; higher awareness of govt scholarships.



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SUSTAINABILITY AND EXPANSION:

- â€¢ State Adoption: Direct deployment with DTE Rajasthan and REAP counselling portal.
- â€¢ Scalability: Extendable to ITI, Medical (NEET), and University admissions across other Indian states.

- Potential impact on the target audience
- Benefits of the solution (social, economic, environmental, etc.)

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RESEARCH AND REFERENCES



- Details / Links of the reference and research work